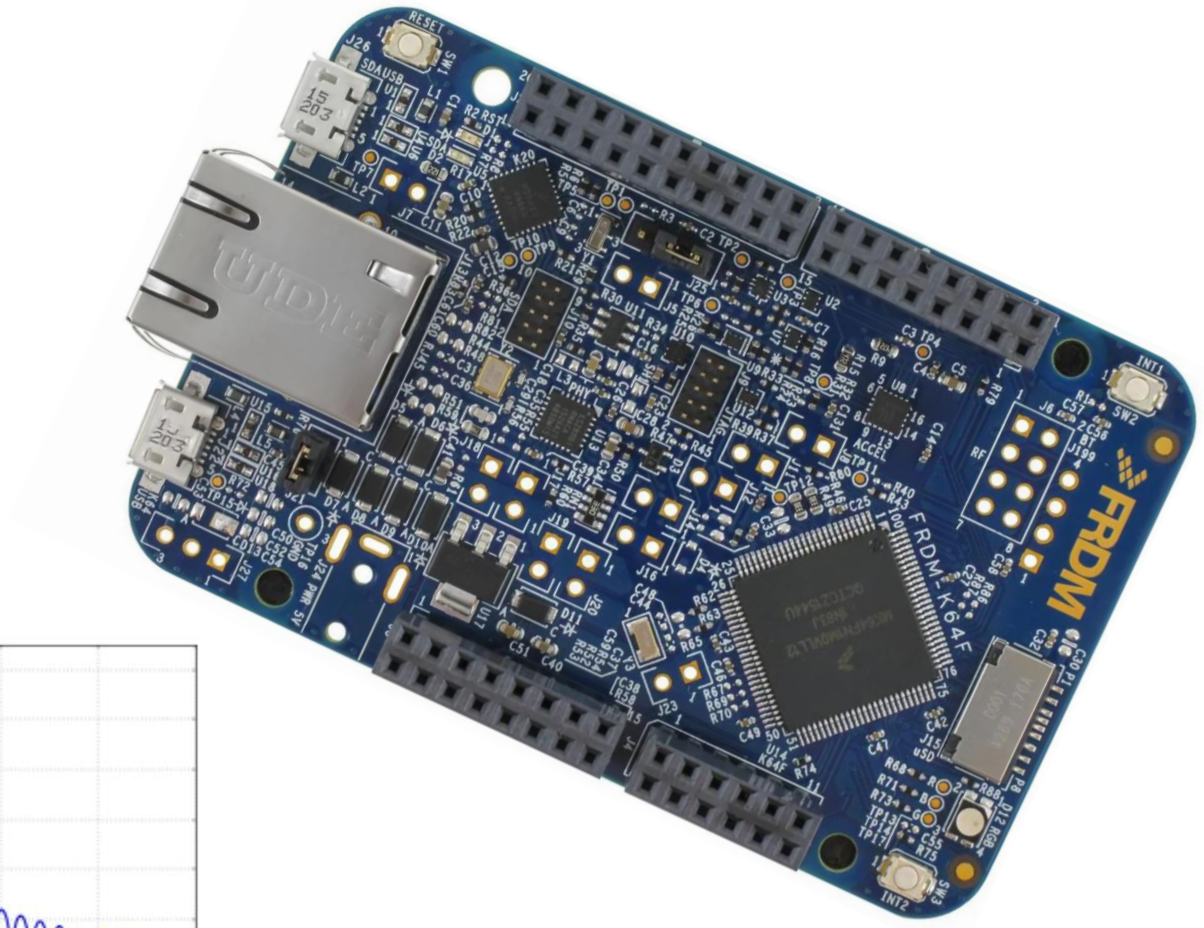
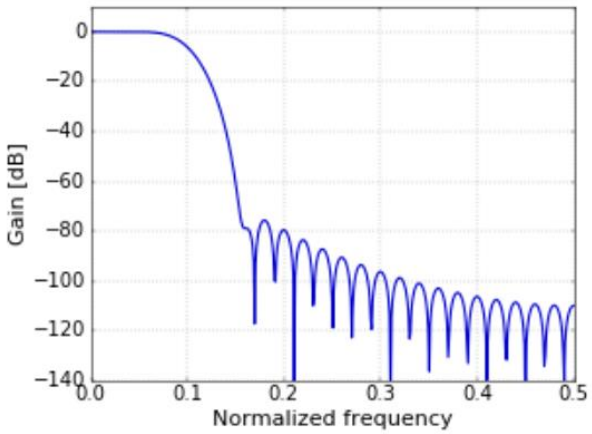
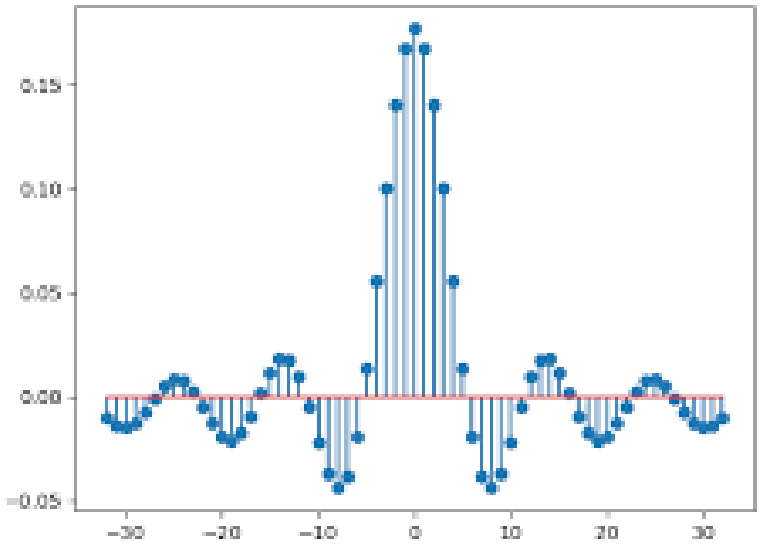
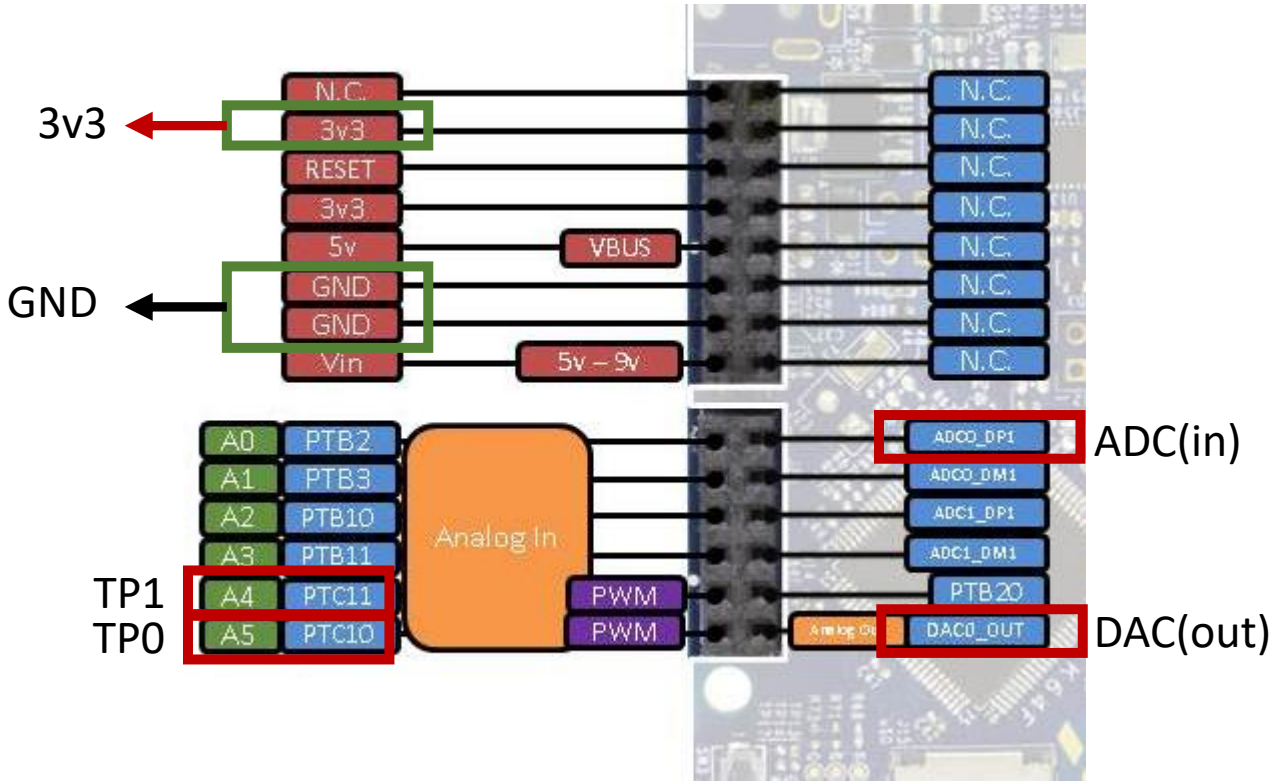
Implementación de filtros digitales

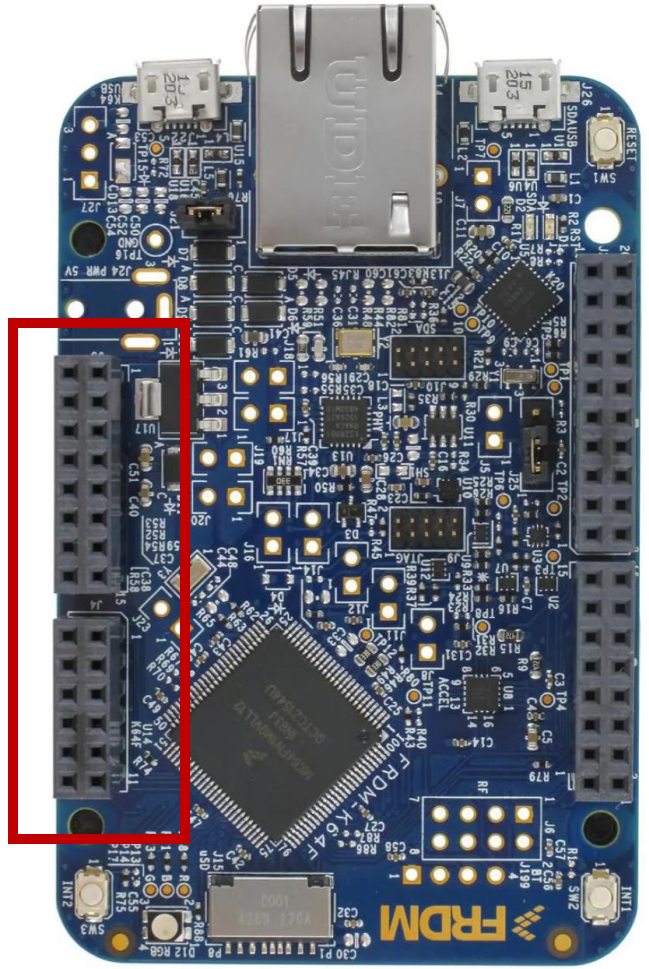






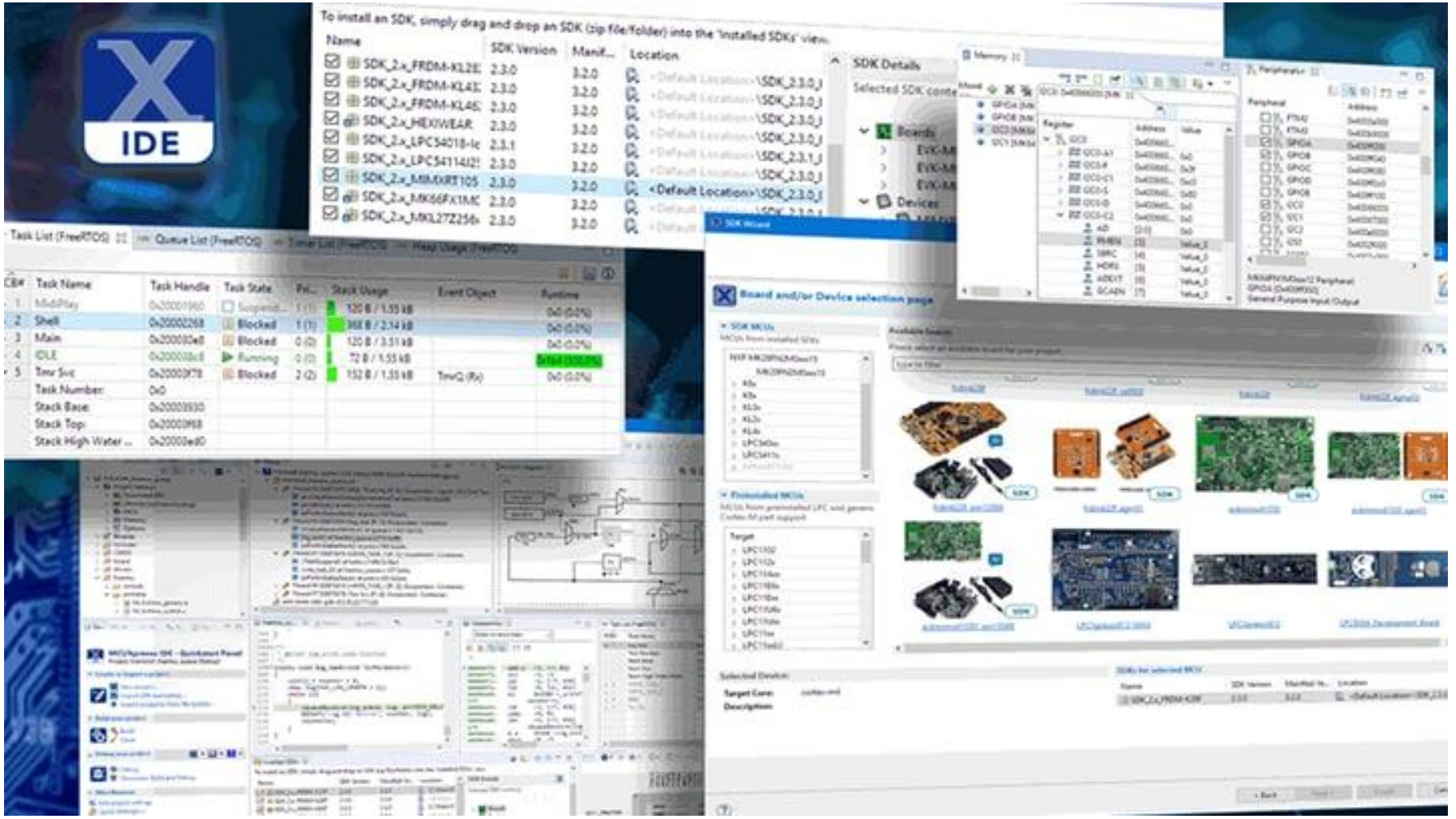
Hardware: FRDM-K64F The Freedom-K64F Link to info





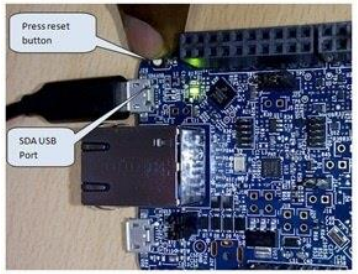
TP0,TP1 Test Points

MCUXpresso IDE [Download]



Setting Open SDA **(Only First Time—New boards)**

1-Connect the board to the PC via SDA USB while keeping the reset button of the board pressed.



2-The board enumerates as BOOTLOADER as shown :

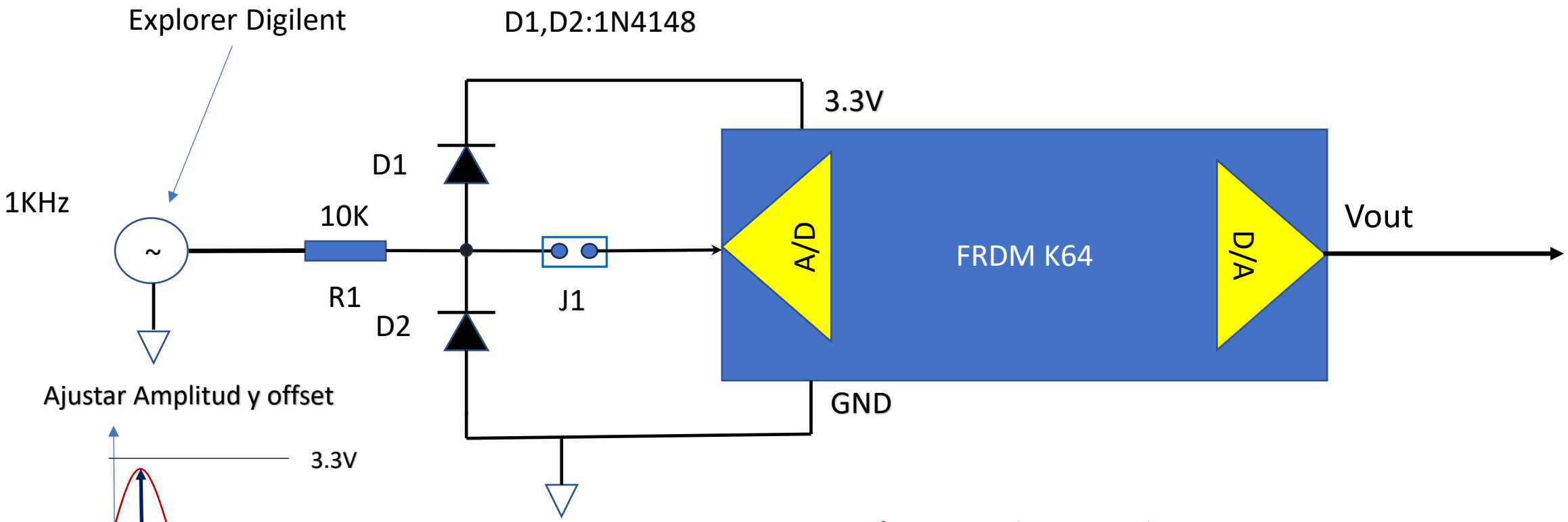


3-Copy the file "**DEBUG-FRDM-K64F_Pemicro_v108a_for_OpenSDA_v2.0.bin"**into the **BOOTLOADER** drive (you can drag and drop too).

4- After the file is copied, the green OpenSDAv2 LED will blink faster.

5 -Unplug the board and power it again

Setting up Hardware for filter development

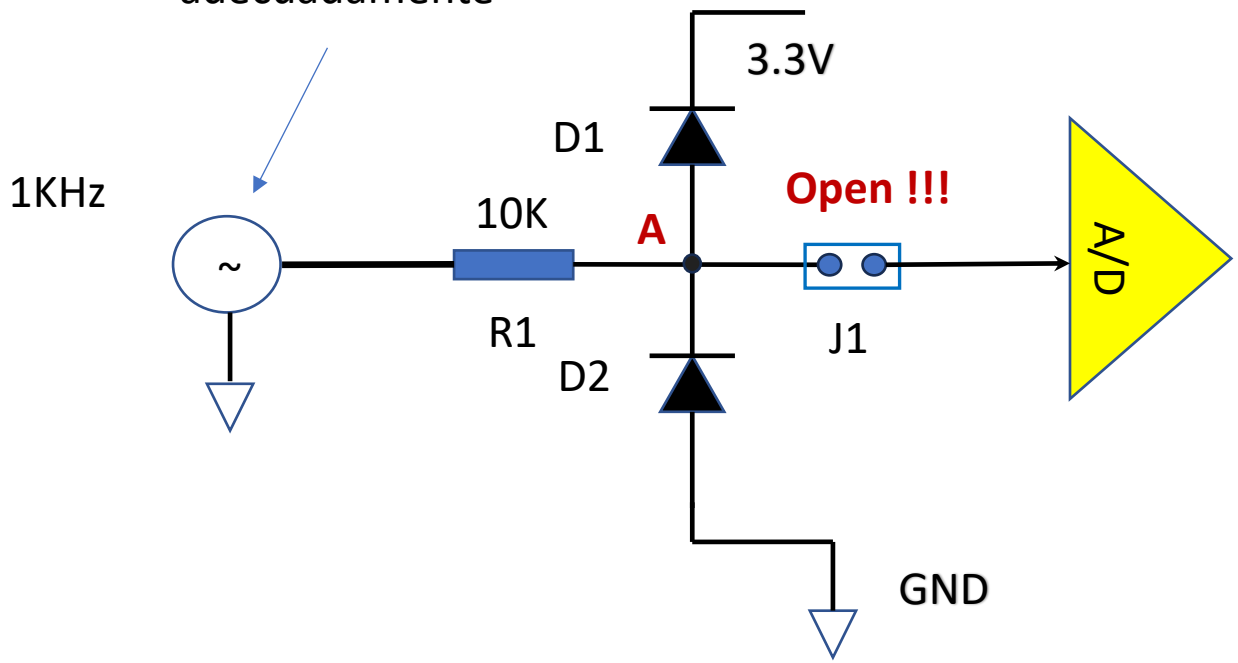


Verificar que el circuito de protección funcione adecuadamente antes de cerrar J1

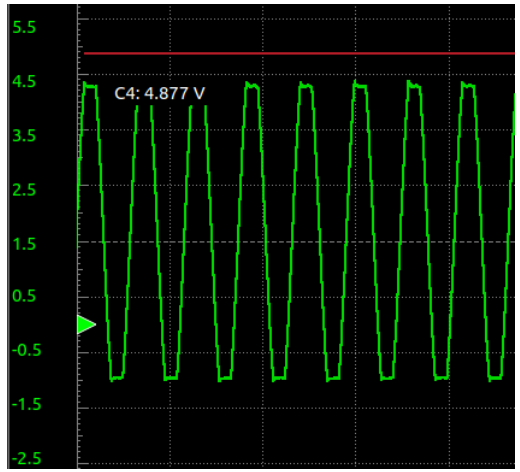
1 – Verificar funcionamiento del circuito de protección (**J1** Abierto !!!)

Aumentar la amplitud para ver si el circuito de protección recorta adecuadamente

D1,D2:1N4148



Nodo A



Verificar que el circuito de protección funcione adecuadamente antes de cerrar J1

2 – Ajuste de amplitud y offset (1/4)

a-Ajustar la amplitud y el offset del generador para que quede como en la Fig. 1 aproximadamente

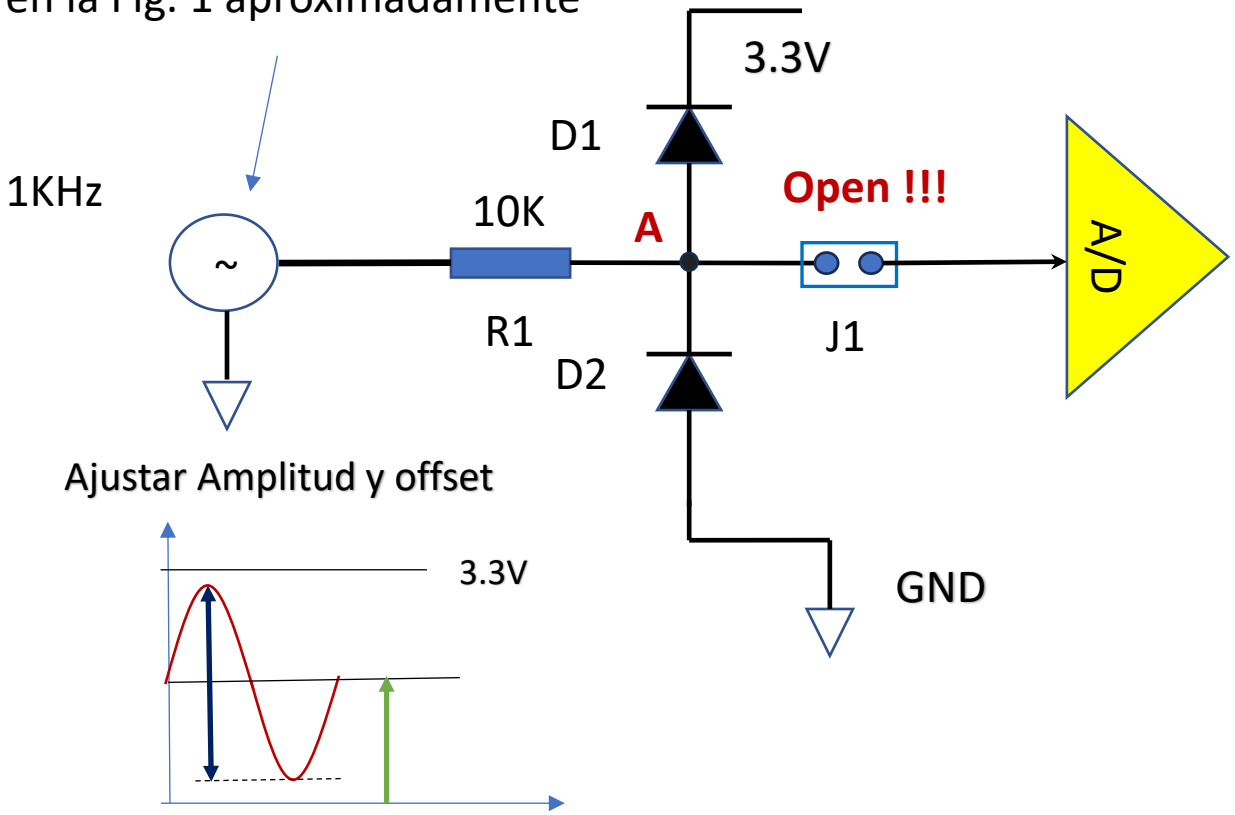
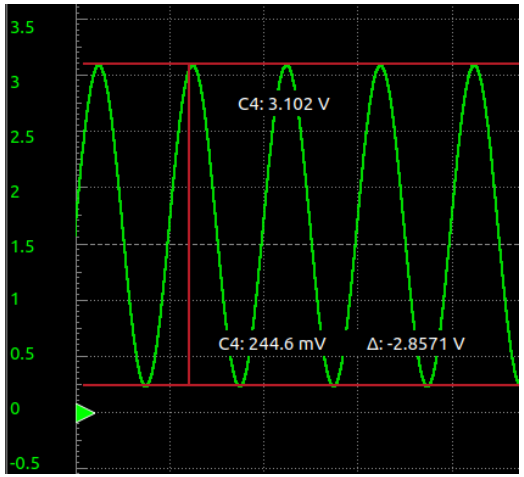


Fig. 1

Nodo A



J1 - Open

2 – Ajuste de amplitud y offset (2/4)

b-Cerrar J1 y ejecutar el modo caño en la Freedom (PIP_T).

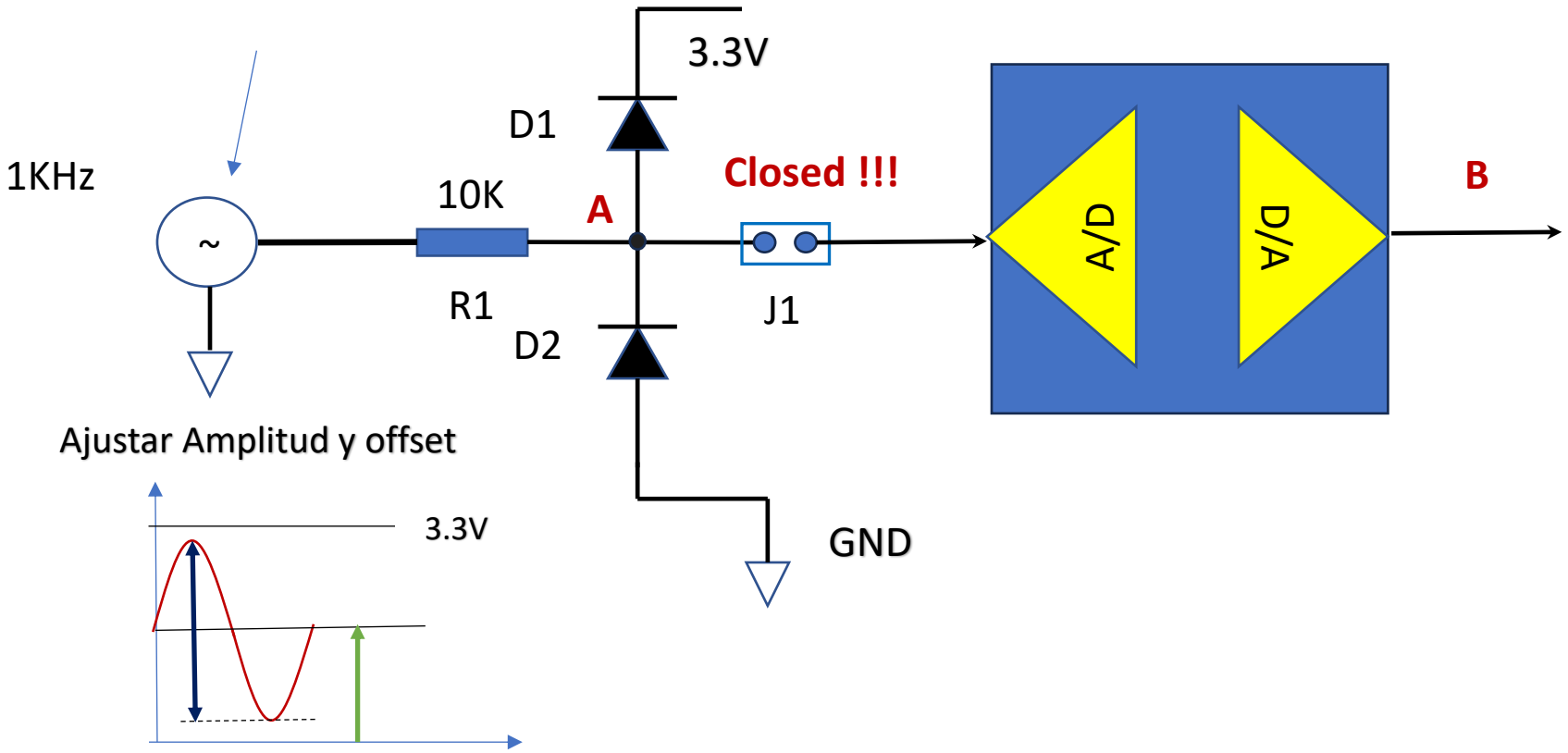
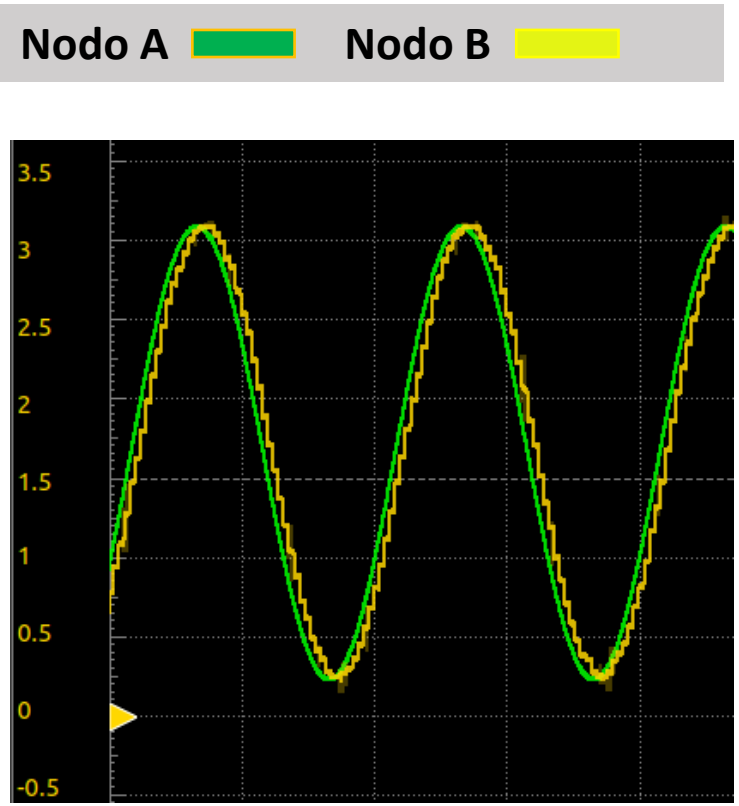
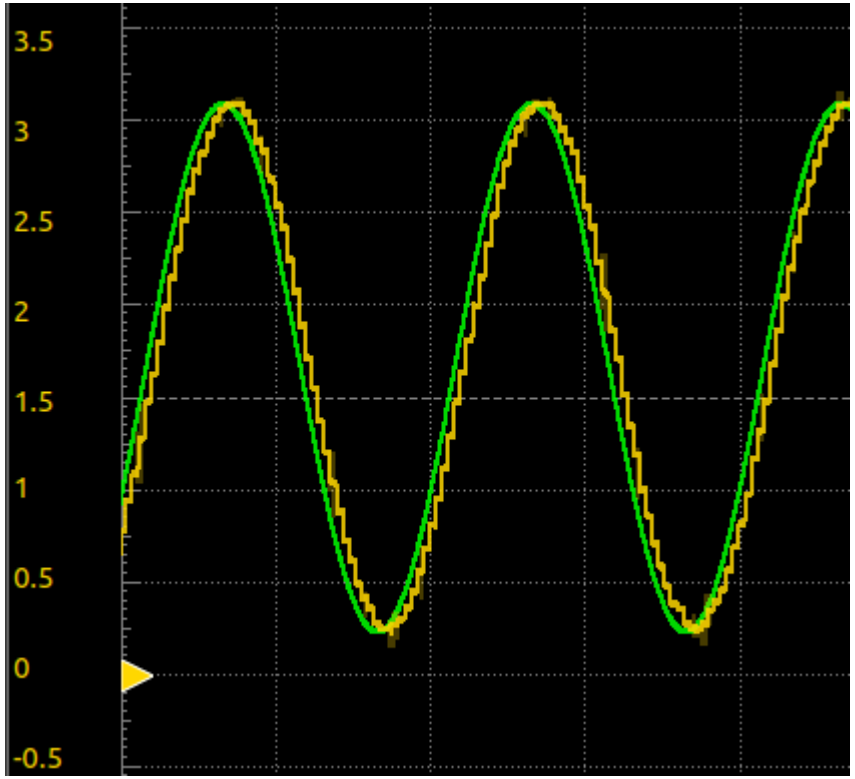


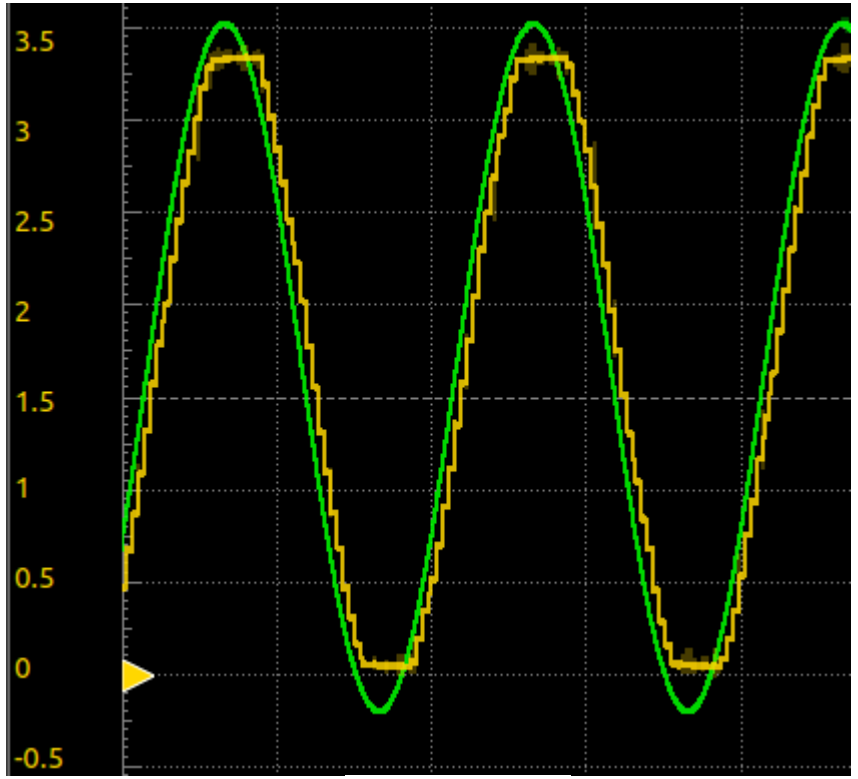
Fig. 1



2 – Ajuste de amplitud y offset (3/4)

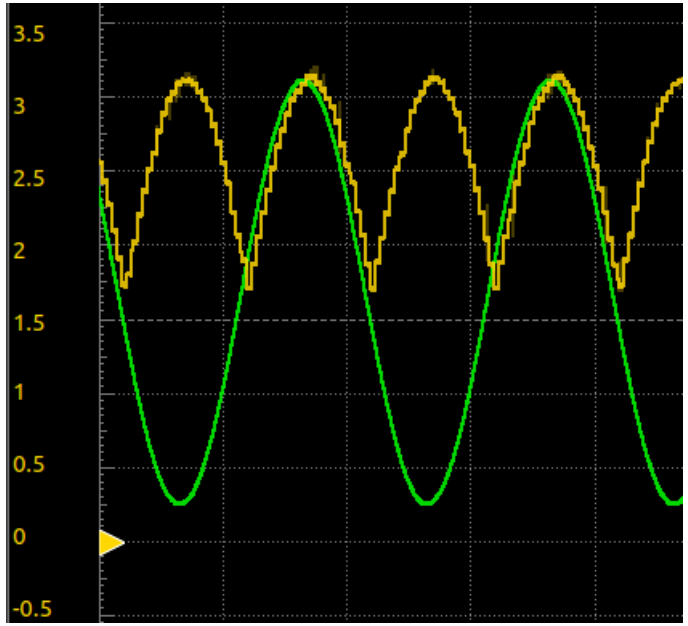
Se puede hacer un ajuste fino de la amplitud asegurándose que no sature la salida (overflow numerico)

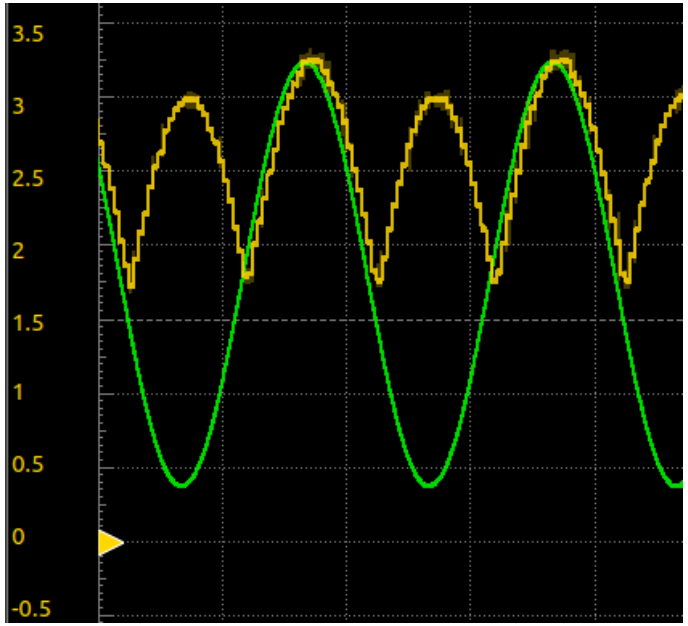




2 – Ajuste de amplitud y offset (4/4)

Una vez ajustada la amplitud se puede hacer un ajuste fino del offset usando el modo calibración (CAL_T) en la freedom. (La salida debe ser simétrica)

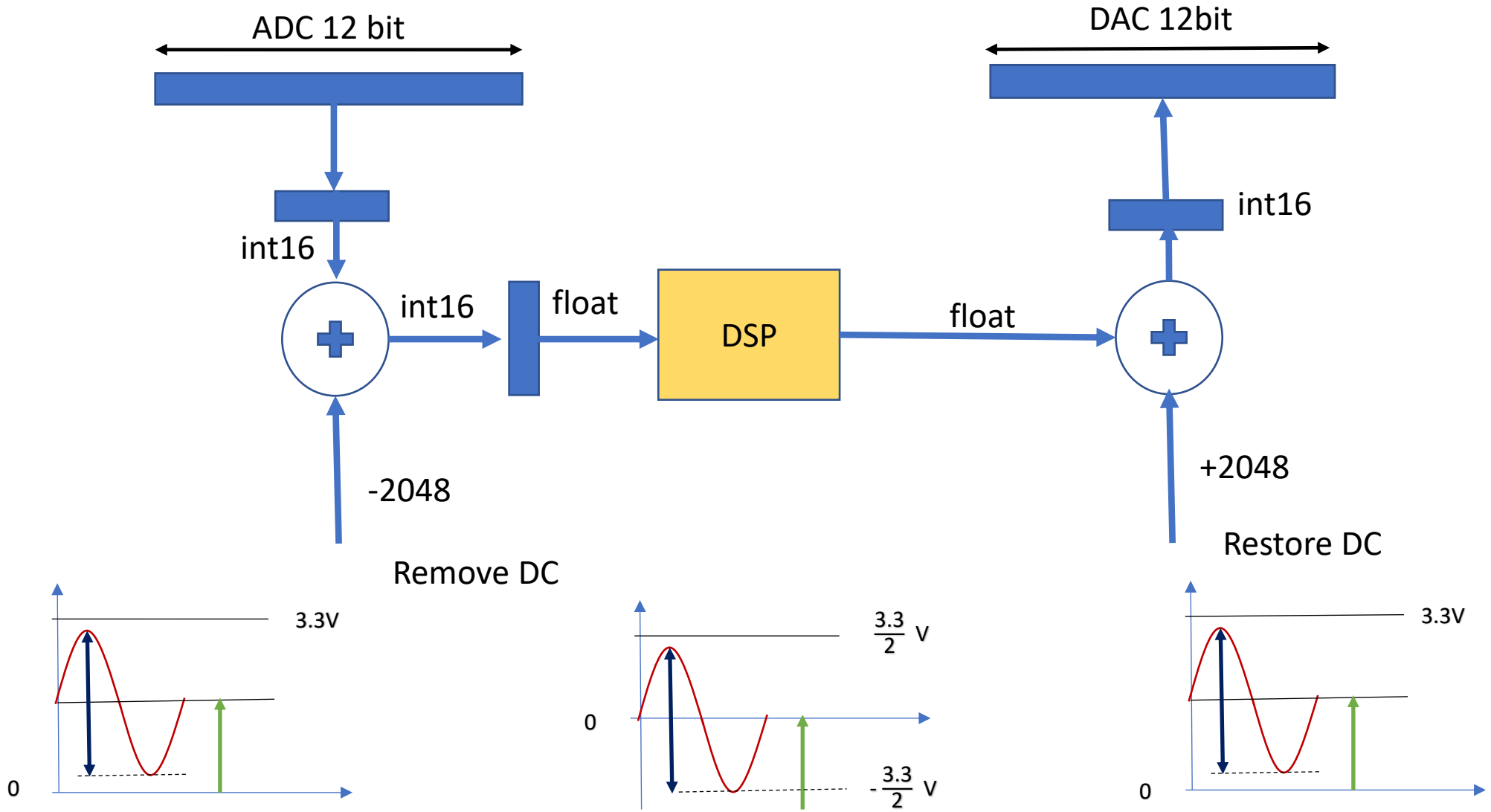








Signal Processing



Los scripts de Matlab para los filtros FIR e IIR generan sus propios headers

